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Inventor(s)	Lin; En-Shuo et al.

Deep trench structure for a capacitive device

Abstract

A deep trench structure may be formed between electrodes of a capacitive device. The deep trench structure may be formed to a depth, a width, and/or an aspect ratio that increases the volume of the deep trench structure relative to a trench structure formed using a metal etch-stop layer. Thus, the deep trench structure is capable of being filled with a greater amount of dielectric material, which increases the capacitance value of the capacitive device. Moreover, the parasitic capacitance of the capacitive device may be decreased by omitting the metal etch-stop layer. Accordingly, the deep trench structure (and the omission of the metal etch-stop layer) may increase the sensitivity of the capacitive device, may increase the humidity-sensing performance of the capacitive device, and/or may increase the performance of devices and/or integrated circuits in which the capacitive device is included.

Inventors: **Lin; En-Shuo (Hsinchu, TW), Ko; Sheng (Hsinchu, TW), Lin; Chi-Fu (Hsinchu County, TW), Lin; Che-Yi (Hsinchu, TW), Lee; Clark (Hsinchu, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.** (Hsinchu, TW)

Family ID: **79327304**

Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)**

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Primary Examiner: Nguyen; Duy T

Attorney, Agent or Firm: Harrity & Harrity, LLP

Background/Summary

RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/949,769, filed on Nov. 13, 2020 (now U.S. Pat. No. 11,658,206), which is incorporated herein by reference in its entirety.

BACKGROUND
(1) Integrated circuits may be fabricated on a semiconductor wafer. Semiconductor wafers can be stacked or bonded on top of each other to form what is referred to as a three-dimensional integrated circuit. Some semiconductor wafers include micro-electromechanical-system (MEMS) devices, which involves the process of forming micro-structures with dimensions in the micrometer scale (one millionth of a meter). Typically, MEMS devices are built on silicon wafers and realized in thin films of materials. Examples of MEMS applications include motion sensors, accelerometers, gyroscopes, and humidity sensors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** is a diagram of an example environment in which systems and/or methods described herein may be implemented.
(3) FIGS. **2**, **3**, **4**A, and **4**B are diagrams of an example capacitive device described herein.
(4) FIGS. **5**A-**5**K are diagrams of an example implementation described herein.
(5) FIG. **6** is a diagram of example components of one or more devices of FIG. **1**.
(6) FIG. **7** is a flowchart of an example process relating to forming a capacitive device described herein.
DETAILED DESCRIPTION
(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(9) A micro-electromechanical-system (MEMS) relative humidity sensor device is a MEMS device

that may include one or more capacitive devices. A capacitive device may be a capacitor or may be a device that includes a plurality of capacitive elements electrically connected in parallel. A dielectric sensor may be placed between electrodes of a capacitor or a capacitive element. The dielectric sensor may be formed of a material having a dielectric constant that changes based on humidity. Changes in humidity cause a change in the dielectric constant of the dielectric sensor, which changes the capacitance of the capacitor or the capacitive element. The changes in capacitance can be converted to a measurement of relative humidity.

(10) A dielectric sensor of a capacitor or a capacitive element may be formed in a trench between two structures on which the electrodes are formed. The trench may formed by etching through one or more layers down to a metal etch-stop layer. However, the use of the metal etch-stop layer may reduce the depth of the trench, which reduces the size of the dielectric sensor and decreases the capacitance of the capacitor or the capacitive element. This may reduce the humidity-sensing performance of the capacitor or the capacitive element. Moreover, parasitic capacitance resulting from the conductivity of the metal etch-stop layer may further reduce the humidity-sensing performance of the capacitor or the capacitive element.

(11) Some implementations described herein provide a deep trench structure for a capacitive device. In some implementations, one or more metal etch-stop layers may be omitted from the capacitive device such that the deep trench structure may be formed between electrodes of the capacitive device down to (and partially in) an interlayer dielectric (ILD) layer of the capacitive device. In this way, the deep trench structure may be formed to a depth and/or an aspect ratio that increases the volume of the deep trench structure relative to a trench structure formed using a metal etch-stop layer. Thus, the deep trench structure is capable of being filled with a greater amount of dielectric material, which increases the capacitance value of the capacitive device. Moreover, the parasitic capacitance of the capacitive device may be decreased by omitting the metal etch-stop layer. Accordingly, the deep trench structure (and the omission of the metal etch-stop layer) may increase the sensitivity of the capacitive device, may increase the humidity-sensing performance of the capacitive device, and/or may increase the performance of devices (e.g., MEMS devices and/or other types of semiconductor devices) and/or integrated circuits in which the capacitive device is included.

(12) FIG. **1** is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. **1**, environment **100** may include a plurality of semiconductor processing tools **102-112**. The plurality of semiconductor processing tools **102-112** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etching tool **108**, a plating tool **110**, a wafer/die transport tool **112**, and/or another type of semiconductor processing tool. The plurality of semiconductor processing tools **102-112** included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing and/or manufacturing facility, and/or the like.

(13) The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

(14) The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV

light source, an extreme UV light source, and/or the like), an x-ray source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

(15) The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

(16) The etching tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** may etch one or more portions of a the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotropically or directionally etch the one or more portions.

(17) The plating tool **110** is a semiconductor processing tool that is capable of plating a substrate (e.g., a wafer, a semiconductor device, and/or the like) or a portion thereof with one or more metals. For example, the plating tool **110** may include a copper electroplating device, an aluminum electroplating device, a nickel electroplating device, a tin electroplating device, a compound material or alloy (e.g., tin-silver, tin-lead, and/or the like) electroplating device, and/or an electroplating device for one or more other types of conductive materials, metals, and/or the like.

(18) Wafer/die transport tool **112** includes a mobile robot, a robot arm, a tram or rail car, an overhead hoist transfer (OHT) vehicle, and/or another type of device that are used to transport wafers and/or dies between semiconductor processing tools **102-110** and/or to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **112** may be a programmed device to travel a particular path and/or may operate semi-autonomously or autonomously.

(19) The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of environment **100** may perform one or more functions described as being performed by another set of devices of environment **100**.

(20) FIGS. **2**, **3**, **4**A, and **4**B are diagrams of an example capacitive device **200**. Capacitive device **200** may be a capacitor or a device that includes a plurality of capacitive elements. In some implementations, capacitive device **200** may be included in another device or system, such as a MEMS device (e.g., a MEMS relative humidity sensor) or an integrated circuit, among other examples.

(21) FIG. **2** shows a perspective view of the example capacitive device **200**. As shown in FIG. **2**, the capacitive device **200** may include a substrate **202**. The substrate **202** may include a semiconductor die substrate, a semiconductor wafer, or another type of substrate in which semiconductor devices may be formed. In some implementations, the substrate **202** is formed of

silicon, a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), a silicon on insulator (SOI), or another type of semiconductor material.
(22) As further shown in FIG. **2**, the capacitive device **200** may include a first dielectric layer **204** above and/or on the substrate **202**. The dielectric layer **204** may be an interlayer dielectric (ILD) layer formed of an electrically insulating material that electrically insulates one or more structures or layers of the capacitive device **200** from other structures or layers of the capacitive device **200**. For example, the dielectric layer **204** may include tantalum nitride (TaN), silicon oxide (SiOx), silicate glass, silicon oxycarbide, a silicon nitride (Si.sub.xN.sub.y), and/or the like.
(23) As further shown in FIG. **2**, the capacitive device **200** may include a dielectric layer **206** above and/or on the dielectric layer **204**. The dielectric layer **206** may be an intermetal dielectric (IMD) layer formed of an electrically insulating material that electrically insulates one or more structures or layers of the capacitive device **200** from one or more metallization layers or metal structures of the capacitive device **200**. For example, the dielectric layer **206** may include tantalum nitride (TaN), silicon oxide (SiOx), silicate glass, silicon oxycarbide, a silicon nitride (Si.sub.xN.sub.y), and/or the like.
(24) As further shown in FIG. **2**, the capacitive device **200** may include a plurality of electrode structures **208** (e.g., electrode structure **208***a* and electrode structure **208***b*). The electrode structures **208** may be formed above and/or on the dielectric layer **206**. Each electrode structure **208** may be formed of a conductive metal capable of carrying an electric charge such as gold, aluminum, or silver, among other examples. An electrode structure **208** may be configured to store an electric charge. For example, the electrode structure **208***a* may be configured to store a positive charge (and thus, may be referred to as a positive charge electrode structure, a positive electrode structure, or a p-electrode structure), and the electrode structure **208***b* may be configured to store a negative charge (and thus, may be referred to as a negative charge electrode structure, a negative electrode structure, or an n-electrode structure).
(25) Each electrode structure **208** may include an electrode pad **210** that electrically connects the electrode structure **208** to interconnects, vias, external contact pads, and/or other structures of the capacitive device **200** (or the device or system in which the capacitive device **200** is included). The electrode pad **210** may connect to a main structure **212**, which may also be referred to as a trunk line, a backbone, and/or the like. A plurality of electrodes **214** may branch off of the main structure **212**.
(26) As further shown in FIG. **2**. the main structure **212** and the electrodes **214** may form a comb structure in which the electrodes **214** are positioned or configured substantially perpendicular to the main structure **212**. Moreover, the electrodes **214** of the electrode structure **208***a* and the electrodes **214** of the electrode structure **208***b* may be interdigitated. In these examples, the electrodes **214** of the electrode structure **208***a* may be positioned or configured in the spaces between the electrodes **214** of the electrode structure **208***b*, and the electrodes **214** of the electrode structure **208***b* may be positioned or configured in the spaces between the electrodes **214** of the electrode structure **208***a*.
(27) As further shown in FIG. **2**, the capacitive device **200** may include a dielectric layer **216** above and/or on the dielectric layer **206**, and in between electrodes **214** of the electrode structures **208**. The dielectric layer **216** may be formed of a dielectric material that is sensitive to humidity, such as a polyimide layer or another polymer that is electrically insulating and sensitive to atmospheric humidity. The dielectric layer **216** may be sensitive to humidity in that the dielectric constant of the dielectric layer **216** changes based on the humidity of the environment in which the capacitive device **200** is located.
(28) The dielectric layer **216** may be located or positioned in a non-conductive region **218** between respective pairs of electrodes **214**. Each pair of electrodes **214** may include an electrode **214** of the electrode structure **208***a* and an electrode **214** of the electrode structure **208***b*. Thus, each pair of electrodes **214** may include a positive electrode (or positive charge electrode or p-electrode) and a negative electrode (or negative charge electrode or n-electrode). Accordingly, when the capacitive

device **200** is in operation, an electric field may be generated in the dielectric layer **216** in a non-conductive region **218** between a pair of electrodes **214** as a result of positive charge stored by a positive electrode and negative charge stored by a negative electrode of the pair of electrodes **214**. A combination of a positive electrode, a negative electrode, and the dielectric layer **216** in a non-conductive region **218** between the positive electrode and the negative electrode may form a capacitive element **220** (or capacitor) of the capacitive device **200**.
(29) FIG. **3** shows a cross-sectional view of a portion of the capacitive device **200** along line AA of FIG. **2**. As shown in FIG. **3**, the dielectric layer **204** may be located above and/or on the substrate **202**, the dielectric layer **206** may be located above and/or on the dielectric layer **204** without an intervening metallization layer between the dielectric layer **204** and the dielectric layer **206**, and the electrode pads **210** and the electrodes **214** of the electrode structures **208** may be located above and/or on the dielectric layer **206**.
(30) As further shown in FIG. **3**, deep trench structures **302** may be formed in the dielectric layer **206** between electrodes **214** such that the electrodes **214** are positioned on substantially trapezoidal shaped structures of the dielectric layer **206**. In particular, a deep trench structure **302** may be located in a non-conductive region **218** between a pair of electrodes **214** configured to store opposing charge of a capacitive element **220**. For example, a deep trench structure **302** may be located between an electrode **214***a* (e.g., configured to store a positive charge or a negative charge) and an electrode **214***b* (e.g., configured to store a type of charge that opposes the type of charge stored by electrode **214***a* so if electrode **214***a* stores a positive charge, electrode **214***b* stores a negative charge, or vice versa). The deep trench structures **302** of the capacitive device **200** may be filled with the dielectric layer **216**. The dielectric layer **216** may also be formed above and/or on the electrodes **214** to protect the electrodes from corrosion and other environmental effects.
(31) In situations where the capacitive device **200** is included in a relative humidity sensing device (e.g., a MEMS relative humidity sensor device or another type of relative humidity sensing device), the dielectric layer **206** in the deep trench structure **302** may function as a humidity sensing layer. In these situations, the dielectric constant of the humidity sensing layer may be based on and/or may change based on humidity in the environment in which the relative humidity sensing device is located. Changes in the dielectric constant of the humidity sensing layer may result in changes to the electric field (and thus, the capacitance) between the electrodes **214***a* and **214***b*. The relative humidity sensing device may include additional circuitry and/or components to measure the electric field and/or the capacitance between the electrodes **214***a* and **214***b* and/or convert the measurement to a relative humidity value.
(32) FIG. **4**A shows a cross-sectional close-up view **304** from a portion of the capacitive device **200** shown in FIG. **3**. As shown in FIG. **4**A, the dielectric layer **204** may be located above and/or on the substrate **202**, the dielectric layer **206** may be located above and/or on the dielectric layer **204** without an intervening metallization layer between the dielectric layer **204** and the dielectric layer **206**, the electrode pads **210** and the electrodes **214***a* and **214***b* of the capacitive element **220** may be located above and/or on the dielectric layer **206**, and the deep trench structure **302** may be located in and/or through the dielectric layer **206** between the electrodes **214***a* and **214***b*.
(33) As further shown in FIG. **4**A, the capacitive device **200** may include one or more passivation layers, such as a passivation layer **402**, a passivation layer **404**, and a passivation layer **406**. The passivation layer **402** may be located above and/or on the electrodes **214***a* and **214***b*. The passivation layer **402** may include an oxide material, such as a silicon oxide (SiO.sub.x), a metallized oxide, or another type of oxide material. The passivation layer **402** may provide outside circuit passivation and may electrically isolate the electrodes **214***a* and **214***b* from other electrodes **214** and other circuits and/or devices of the capacitive device **200**.
(34) The passivation layer **404** may be located above and/or on passivation layer **402** (e.g., which is above and/or on the electrodes **214***a* and **214***b*). The passivation layer **404** may include a nitride material, such as a silicon nitride (Si.sub.xN.sub.y) or another type of nitride material. The

passivation layer **404** may provide outside circuit passivation and may electrically isolate the electrodes **214***a* and **214***b* from other electrodes **214** and other circuits and/or devices of the capacitive device **200**.

(35) The passivation layer **406** may be located above and/or on passivation layer **404** (e.g., which is above and/or on the electrodes **214***a* and **214***b*). Moreover, the passivation layer **406** may be located in the deep trench structure **302**. In particular, the passivation layer **406** may be located on the bottom of the deep trench structure **302** and on the sidewalls of the deep trench structure **302**. In this way, the passivation layer **406** in the deep trench structure **302** forms a trench liner that provides cavity passivation for the deep trench structure **302**. In some implementations, the passivation layer **406** includes a nitride material such as a silicon nitride (Si.sub.xN.sub.y) or another type of nitride material. The dielectric layer **216** (e.g., the humidity sensing layer, in some implementations) may be located above and/or on the passivation layer **406** in the deep trench structure **302** and above the electrodes **214***a* and **214***b*.

(36) As further shown in FIG. **4**A, the deep trench structure **302** may be located in a portion of the dielectric layer **204** below a top surface of the dielectric layer **204** referred to as an over-etch region **408**. The over-etch region **408** may be formed in the dielectric layer **204** during etching of the dielectric layer **206** when forming the deep trench structure **302** to achieve a particular trench depth of the deep trench structure **302**, to achieve a particular trench width of the deep trench structure **302**, and/or to achieve a particular aspect ratio for the deep trench structure **302**. In these examples, the bottom of the deep trench structure **302** is located in the over-etch region **408**, and therefore is located in a portion of the dielectric layer **204** below the top surface of the dielectric layer **204**. The passivation layer **406** may be formed on the dielectric layer **204** in the over-etch region **408** at the bottom of the deep trench structure **302** and at least partially below the top surface of the dielectric layer **204**.

(37) FIG. **4**B shows a cross-sectional close-up view **410** from a portion of the capacitive device **200** shown in FIG. **4**A. As shown in FIG. **4**B, various layers and/or structures of the capacitive device **200** may be formed to particular dimensions or dimensional ranges. In particular, the dielectric layer **206** may be formed to a height (or thickness) a, the electrodes **214***a* and **214***b* may be formed to a height (or thickness) b, the passivation layer **402** may be formed to a height (or thickness) c, the passivation layer **404** may be formed to a height (or thickness) d, and/or the passivation layer **404** may be formed to a height (or thickness) e. The dielectric layer **206** may be formed to the height a such that a particular depth f of the deep trench structure **302** may be achieved, such that a particular aspect ratio of the deep trench structure **302** may be achieved, and/or such that the volume within the deep trench structure **302** may be achieved. As an example, the height a of the dielectric layer **206** may be approximately 24,000 angstroms.

(38) The electrodes **214***a* and **214***b* may each be formed to the height b such that a particular charge-storage capacity of the electrodes **214***a* and **214***b* may be achieved, such that a particular capacitance value for the capacitive device **200** and/or the capacitive element **220** may be achieved, and/or such that a particular capacitance value range for the capacitive device **200** and/or the capacitive element **220** may be achieved. As an example, the height b of the electrodes **214***a* and **214***b* may be approximately 8,000 angstroms.

(39) The passivation layer **402** may be formed to the height c such that the passivation layer **402** may provide a particular amount of circuit passivation, such that a particular depth f of the deep trench structure **302** may be achieved, such that a particular aspect ratio of the deep trench structure **302** may be achieved, and/or such that the volume within the deep trench structure **302** may be achieved. As an example, the height c of the passivation layer **402** may be approximately 2,000 angstroms.

(40) The passivation layer **404** may be formed to the height d such that the passivation layer **404** may provide a particular amount of circuit passivation, such that a particular depth f of the deep trench structure **302** may be achieved, such that a particular aspect ratio of the deep trench structure

302 may be achieved, and/or such that the volume within the deep trench structure **302** may be achieved. As an example, the height d of the passivation layer may be approximately 3,000 angstroms.

(41) The passivation layer **406** may be formed to the height e such that the passivation layer **406** may provide a particular amount of trench passivation, such that a particular depth f of the deep trench structure **302** may be achieved, such that a particular aspect ratio of the deep trench structure **302** may be achieved, and/or such that the volume within the deep trench structure **302** may be achieved. As an example, the height e of the passivation layer may be approximately 4,000 angstroms.

(42) The over-etch region **408** may be formed to the depth g such that a particular depth f of the deep trench structure **302** may be achieved, such that a particular aspect ratio of the deep trench structure **302** may be achieved, and/or such that the volume within the deep trench structure **302** may be achieved. As an example, the depth g of the over-etch region **408** may be in a range of approximately 1,000 angstroms to approximately 9,000 angstroms.

(43) The deep trench structure **302** may be formed to the depth f, the width h, the sidewall angle j, and/or to a particular aspect ratio between the width h and the depth f such that one or more operational parameters and/or performance parameters for the capacitive device **200** and/or the capacitive element **220** are achieved. As an example, the deep trench structure **302** may be formed to the depth f, the width h, the sidewall angle j, and/or to a particular aspect ratio between the width h and the depth f such that a particular capacitive value or capacitive value range (e.g., approximately 15,920 picofarads, approximately 15,650 picofarads to approximately 16,060 picofarads, among other examples) for the capacitive device **200** and/or the capacitive element **220** is achieved. As another example, the deep trench structure **302** may be formed to the depth f, the width h, the sidewall angle j, and/or to a particular aspect ratio between the width h and the depth f such that a threshold amount of parasitic capacitive for the capacitive device **200** and/or the capacitive element **220** is achieved. As another example, the deep trench structure **302** may be formed to the depth f, the width h, the sidewall angle j, and/or to a particular aspect ratio between the width h and the depth f such that a particular amount of volume (e.g., an amount of volume in which the dielectric layer **216** may be deposited) in the deep trench structure **302** is achieved.

(44) An example depth f of the deep trench structure **302** may be in a range of approximately 42,000 angstroms to approximately 50,000 angstroms to achieve and/or satisfy one or more of the operational parameters and/or performance parameters described above. An example aspect ratio of the deep trench structure **302**, between the width h of the deep trench structure **302** and the depth f of the deep trench structure **302**, may be in a range of approximately 0.26 to approximately 0.38 to achieve and/or satisfy one or more of the operational parameters and/or the performance parameters described above. In some implementations, an example width h of the deep trench structure **302** is in a range of approximately 13,000 angstroms to approximately 15,000 angstroms to achieve and/or satisfy one or more of the operational parameters and/or performance parameters described above. In some implementations, an example width h of the deep trench structure **302** is greater than approximately 15,000 angstroms to achieve and/or satisfy one or more of the operational parameters and/or performance parameters described above. An example sidewall angle j of the sidewalls of the deep trench structure **302** may be in a range of approximately 7 degrees to approximately 8 degrees to achieve and/or satisfy one or more of the operational parameters and/or performance parameters described above.

(45) The electrodes **214***a* and **214***b* may each be formed to a width k such that a particular charge-storage capacity of the electrodes **214***a* and **214***b* may be achieved, such that a particular capacitance value for the capacitive device **200** and/or the capacitive element **220** may be achieved, and/or such that a particular capacitance value range for the capacitive device **200** and/or the capacitive element **220** may be achieved. As an example, the width k of the electrodes **214***a* and **214***b* may be approximately 11,000 angstroms.

(46) As indicated above, FIGS. **2**, **3**, **4**A, and **4**B are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. **2**, **3**, **4**A, and **4**B.
(47) FIGS. **5**A-**5**K are diagrams of an example implementation **500** described herein. In particular, example implementation **500** may be an example of forming the capacitive device **200** or a portion thereof. As shown in FIG. **5**A, the portion of the capacitive device **200** may include a capacitive element **220**. As further shown in FIG. **5**A, the capacitive device **200** may include the substrate **202** on which other layers and/or structures of the capacitive device **200** may be formed.
(48) As shown in FIG. **5**B, the dielectric layer **204** (e.g., the ILD layer) may be formed above and/or on the substrate **202**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the dielectric layer **204** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique.
(49) As shown in FIG. **5**C, a first portion **206***a* of the dielectric layer **206** (e.g., the IMD layer) may be formed above and/or on the dielectric layer **204**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the first portion **206***a* of the dielectric layer **206** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique.
(50) As shown in FIG. **5**D, a second portion **206***b* of the dielectric layer **206** (e.g., the IMD layer) may be formed above and/or on the first portion **206***a* of the dielectric layer **206**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the second portion **206***b* of the dielectric layer **206** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique.
(51) In some implementations, the dielectric layer **206** is composed of the first portion **206***a* and the second portion **206***b*, and the first portion **206***a* and the second portion **206***b* are formed in separate deposition operations. In some implementations, the height (or thickness) of the first portion **206***a* and the height (or thickness) of the second portion **206***b* are the same height (or thickness). In some implementations, the height (or thickness) of the first portion **206***a* and the height (or thickness) of the second portion **206***b* are different heights (or different thicknesses). In some implementations, the dielectric layer **206** is composed of a single dielectric layer that is formed in a single deposition operation. The dielectric layer **206** may be formed above and/or on the dielectric layer **204** without an intervening metallization layer between the dielectric layer **204** and the dielectric layer **206**.
(52) As shown in FIG. **5**E, a metallization layer **502** may be formed above and/or on the dielectric layer **206**. A semiconductor processing tool may form or deposit the metallization layer **502** above and/or on the dielectric layer **206**. In some implementations, the deposition tool **102** deposits the metallization layer **502** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique. In some implementations, the plating tool **110** deposits the metallization layer **502** using a plating technique such as electroplating (or electro-chemical deposition). In these examples, the plating tool **110** may apply a voltage across an anode formed of a plating material and a cathode (e.g., a substrate). The voltage causes a current to oxidize the anode, which causes the release of plating material ions from the anode. These plating material ions form a plating solution that travels through a plating bath toward the capacitive device **200**. The plating solution reaches the capacitive device **200** and deposits plating material ions onto the dielectric layer **206** to form the metallization layer **502**.
(53) As shown in FIG. **5**F, a plurality of portions of the metallization layer **502** may be etched through to the dielectric layer **206** to form the electrode pads **210** and the electrodes **214** of the electrode structures **208** included in the capacitive device **200**. For example, an electrode pad **210** and one or more electrodes **214***a* may be formed for an electrode structure **208***a*, and another electrode pad **210** and one or more electrodes **214***b* may be formed for an electrode structure **208***b*. The electrode pads **210** and the electrodes **214** may be formed by coating the metallization layer **502** with a photoresist (e.g., using the deposition tool **102**), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool **104**), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool

106), and etching the plurality of portions of the metallization layer **502** to the dielectric layer **206** based on the pattern in the photoresist. In some implementations, the metallization layer **502** may be formed to a height (or thickness) such that the electrodes **214***a* and **214***b* satisfy a capacitance value parameter for the capacitive device **200**, and/or the metallization layer **502** may be etched so that a width of the electrodes **214***a* and **214***b* satisfy the capacitance value parameter for the capacitive device **200**.

(54) As shown in FIG. **5**G, a deep trench structure **302** may be formed in and/or through the metallization layer **502**, and in and/or through the dielectric layer **206**. Moreover, the deep trench structure **302** may be formed at least partially in and/or at least partially through the dielectric layer **204** such that an over-etch region **408** is formed below the top surface of the dielectric layer **204**. The deep trench structure **302** may be formed by coating the metallization layer **502** and/or the dielectric layer **206** with a photoresist (e.g., using the deposition tool **102**), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool **104**), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool **106**), and etching through the dielectric layer **206** and a portion of the dielectric layer **204** based on the pattern in the photoresist to form the deep trench structure **302** and the over-etch region **408**. In some implementations, the deep trench structure **302** may be formed to a depth, a width, an aspect ratio, and/or a sidewall angle such that the deep trench structure **302** satisfies a capacitance value parameter for the capacitive device **200**, such that the deep trench structure **302** satisfies a parasitic capacitance parameter for the capacitive device **200**, such that a particular volume of dielectric material can be filled in the deep trench structure **302**, and/or such that other operation parameters and/or performance parameters of the capacitive device **200** are achieved and/or satisfied.

(55) As shown in FIG. **5**H, a passivation layer **402** (e.g., a circuit passivation layer) may be formed above and/or on the electrodes **214***a* and **214***b*. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the passivation layer **402** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique. In some implementations, the passivation layer **402** may be formed to a height (or thickness) to satisfy a circuit passivation parameter for the capacitive device **200**.

(56) As shown in FIG. **5**I, a passivation layer **404** (e.g., a circuit passivation layer) may be formed above and/or on the passivation layer **402**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the passivation layer **404** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique. In some implementations, the passivation layer **404** may be formed to a height (or thickness) to satisfy a circuit passivation parameter for the capacitive device **200**.

(57) As shown in FIG. **5**J, a passivation layer **406** (e.g., a trench passivation layer) may be formed above and/or on the passivation layer **404** and in the deep trench structure **302**. In particular, the passivation layer **406** may be formed on the bottom of the deep trench structure **302** in the over-etch region **408** and on the sidewalls of the deep trench structure **302**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the passivation layer **406** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique. In some implementations, the passivation layer **406** may be formed to a height (or thickness) to satisfy a trench passivation parameter for the capacitive device **200**.

(58) As shown in FIG. **5**K, a dielectric layer **216** (e.g., a humidity sensing layer) may be formed above and/or on the passivation layer **406**, above and/or on the electrodes **214***a* and **214***b*, and in the deep trench structure **302**. A semiconductor processing tool (e.g., the deposition tool **102**) may deposit the dielectric layer **216** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique. In some implementations, the dielectric layer **216** may be formed to a height (or thickness) and/or of a particular dielectric material to satisfy a humidity sensing capability parameter for the capacitive device **200**, to satisfy a capacitance value parameter

for the capacitive device **200**, to satisfy a parasitic capacitance parameter for the capacitive device **200**, and/or to satisfy other operation parameters and/or performance parameters of the capacitive device **200**.
(59) As indicated above, FIGS. **5**A-**5**K are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. **5**A-**5**K. In some implementations, the process of forming the capacitive device **200** may include additional techniques and/or procedures, fewer techniques and/or procedures, different techniques and/or procedures, or differently arranged techniques and/or procedures than those depicted in FIG. **5**A-**5**K.
(60) FIG. **6** is a diagram of example components of a device **600**. In some implementations, one or more of the semiconductor processing tools **102-112** may include one or more devices **600** and/or one or more components of device **600**. As shown in FIG. **6**, device **600** may include a bus **610**, a processor **620**, a memory **630**, a storage component **640**, an input component **650**, an output component **660**, and a communication component **670**.
(61) Bus **610** includes a component that enables wired and/or wireless communication among the components of device **600**. Processor **620** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **620** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor **620** includes one or more processors capable of being programmed to perform a function. Memory **630** includes a random access memory, a read only memory, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).
(62) Storage component **640** stores information and/or software related to the operation of device **600**. For example, storage component **640** may include a hard disk drive, a magnetic disk drive, an optical disk drive, a solid state disk drive, a compact disc, a digital versatile disc, and/or another type of non-transitory computer-readable medium. Input component **650** enables device **600** to receive input, such as user input and/or sensed inputs. For example, input component **650** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system component, an accelerometer, a gyroscope, and/or an actuator. Output component **660** enables device **600** to provide output, such as via a display, a speaker, and/or one or more light-emitting diodes. Communication component **670** enables device **600** to communicate with other devices, such as via a wired connection and/or a wireless connection. For example, communication component **670** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.
(63) Device **600** may perform one or more processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **630** and/or storage component **640**) may store a set of instructions (e.g., one or more instructions, code, software code, and/or program code) for execution by processor **620**. Processor **620** may execute the set of instructions to perform one or more processes described herein. In some implementations, execution of the set of instructions, by one or more processors **620**, causes the one or more processors **620** and/or the device **600** to perform one or more processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.
(64) The number and arrangement of components shown in FIG. **6** are provided as an example. Device **600** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **6**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **600** may perform one or more functions described as being performed by another set of components of device **600**.
(65) FIG. **7** is a flowchart of an example process **700** associated with forming a capacitive device.

In some implementations, one or more process blocks of FIG. **7** may be performed by one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools **102-112**). Additionally, or alternatively, one or more process blocks of FIG. **7** may be performed by one or more components of device **600**, such as processor **620**, memory **630**, storage component **640**, input component **650**, output component **660**, and/or communication component **670**.

(66) As shown in FIG. **7**, process **700** may include forming a first dielectric layer on a substrate of a capacitive device (block **710**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a first dielectric layer **204** on a substrate **202** of a capacitive device **200**, as described above.

(67) As further shown in FIG. **7**, process **700** may include forming a second dielectric layer on the first dielectric layer (block **720**). For example, a semiconductor processing tool (e.g., the deposition tool **102**) may form a second dielectric layer **206** on the first dielectric layer **204**, as described above.

(68) As further shown in FIG. **7**, process **700** may include forming a metal layer on the second dielectric layer (block **730**). For example, a semiconductor processing tool (e.g., the deposition tool **102**, the plating tool **110**, and/or another semiconductor processing tool) may form a metal layer **502** on the second dielectric layer **206**, as described above.

(69) As further shown in FIG. **7**, process **700** may include etching the metal layer to form a first electrode of the capacitive device, a first electrode pad, associated with the first electrode, of the capacitive device, a second electrode of the capacitive device, and a second electrode pad, associated with the second electrode, of the capacitive device (block **740**). For example, one or more semiconductor processing tools (e.g., the deposition tool **102**, the exposure tool **104**, the developer tool **106**, the etching tool **108**, and/or another semiconductor processing tool) may etch the metal layer **502** to form a first electrode **214***a* of the capacitive device **200**, a first electrode pad **210** associated with the first electrode **214***a* of the capacitive device **200**, a second electrode **214***b* of the capacitive device **200**, and a second electrode pad **210** associated with the second electrode **214***b* of the capacitive device **200**, as described above.

(70) As further shown in FIG. **7**, process **700** may include etching through the second dielectric layer and into a portion of the first dielectric layer to form a deep trench structure between the first electrode and the second electrode (block **750**). For example, one or more semiconductor processing tools (e.g., the deposition tool **102**, the exposure tool **104**, the developer tool **106**, the etching tool **108**, and/or another semiconductor processing tool) may etch through the second dielectric layer **206** and into a portion **408** of the first dielectric layer **204** to form a deep trench structure **302** between the first electrode **214***a* and the second electrode **214***b*, as described above.

(71) As further shown in FIG. **7**, process **700** may include forming a humidity sensing layer in the deep trench structure (block **760**). For example, the semiconductor processing tool (e.g., the deposition tool **102**) may form a humidity sensing layer **216** in the deep trench structure **302**, as described above.

(72) Process **700** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(73) In a first implementation, process **700** includes forming (e.g., using the deposition tool **102**) a first electrical passivation layer **402** on the first electrode **214***a* and on the second electrode **214***b*, forming (e.g., using the deposition tool **102**) a second electrical passivation layer **404** on the first electrical passivation layer **402**, and forming (e.g., using the deposition tool **102**) a trench passivation layer **406** on the second electrical passivation layer **404** and in the deep trench structure **302**. In a second implementation, alone or in combination with the first implementation, forming the humidity sensing layer **216** in the deep trench structure **302** includes forming the humidity sensing layer **216** over the trench passivation layer **406** in the deep trench structure **302**.

(74) In a third implementation, alone or in combination with one or more of the first and second

implementations, etching through the second dielectric layer **206** and into the portion of the first dielectric layer **204** to form the deep trench structure **302** includes etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** to a particular height such that the capacitive device **200** satisfies at least one of a capacitance value parameter or a parasitic capacitance parameter. In a fourth implementation, alone or in combination with one or more of the first through third implementations, etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** includes etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** to a particular width such that the capacitive device **200** satisfies at least one of a capacitance value parameter or a parasitic capacitance parameter.
(75) In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** includes etching through the second dielectric layer **206** and into the portion of the first dielectric layer **204** to form the deep trench structure **302** to a particular aspect ratio such that the capacitive device **200** satisfies at least one of a capacitance value parameter or a parasitic capacitance parameter. In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** includes etching through the second dielectric layer **206** and into the portion **408** of the first dielectric layer **204** to form the deep trench structure **302** to a particular volume such that the capacitive device **200** satisfies at least one of a capacitance value parameter or a parasitic capacitance parameter.
(76) In a seventh implementation, alone or in combination with one or more of the first through sixth implementations, etching the metal layer **502** to form the first electrode **214***a* and the second electrode **214***b* includes etching the metal layer **502** to form the first electrode **214***a* and the second electrode **214***b* to respective widths such that the capacitive device **200** satisfies a capacitance parameter. In an eighth implementation, alone or in combination with one or more of the first through seventh implementations, forming the second dielectric layer **206** on the first dielectric layer **204** includes forming the second dielectric layer **206** directly on the first dielectric layer **204** without an intervening metallization layer.
(77) Although FIG. **7** shows example blocks of process **700**, in some implementations, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **7**. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.
(78) In this way, one or more metal etch-stop layers may be omitted from the capacitive device such that the deep trench structure may be formed between electrodes of the capacitive device down to (and partially in) an ILD layer of the capacitive device. In this way, the deep trench structure may be formed to a depth and/or an aspect ratio that increases the volume of the deep trench structure relative to a trench structure formed using a metal etch-stop layer. Thus, the deep trench structure is capable of being filled with a greater amount of dielectric material, which increases the capacitance value of the capacitive device. Moreover, the parasitic capacitance of the capacitive device may be decreased by omitting the metal etch-stop layer. Accordingly, the deep trench structure (and the omission of the metal etch-stop layer) may increase the sensitivity of the capacitive device, may increase the humidity-sensing performance of the capacitive device, and/or may increase the performance of devices (e.g., MEMS devices and/or other types of semiconductor devices) and/or integrated circuits in which the capacitive device is included.
(79) As described in greater detail above, some implementations described herein provide a capacitive device. The capacitive device includes a first electrode and a second electrode. The capacitive device includes a deep trench structure between the first electrode and the second

electrode. A bottom of the deep trench structure is in an over-etch region that is below a surface of an ILD layer. The ILD layer is below the first electrode and the second electrode. The capacitive device includes a dielectric layer in the deep trench structure.

(80) As described in greater detail above, some implementations described herein provide a capacitive device. The capacitive device includes a positive charge electrode structure including a plurality of positive electrodes connected to a first electrode pad. The capacitive device includes a negative charge electrode structure including a plurality of negative electrodes connected to a second electrode pad. The capacitive device includes a plurality of deep trench structures. A deep trench structure, of the plurality of deep trench structures, is located between a pair of a positive electrode of the plurality of positive electrodes and a negative electrode of the plurality of negative electrodes. An aspect ratio, between a width of the deep trench structure and a height of the deep trench structure, is in a range of approximately 0.26 to approximately 0.38. The capacitive device includes a humidity sensing layer in the plurality of deep trench structures.

(81) As described in greater detail above, some implementations described herein provide a method. The method includes forming a first dielectric layer on a substrate of a capacitive device. The method includes forming a second dielectric layer on the first dielectric layer. The method includes forming a metal layer on the second dielectric layer. The method includes etching the metal layer to form a first electrode of the capacitive device, a first electrode pad, associated with the first electrode, of the capacitive device, a second electrode of the capacitive device, and a second electrode pad, associated with the second electrode, of the capacitive device. The method includes etching through the second dielectric layer and into a portion of the first dielectric layer to form a deep trench structure between the first electrode and the second electrode. The method includes forming a humidity sensing layer in the deep trench structure. The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming an interlayer dielectric (ILD) layer on a substrate of a capacitive device; forming a first portion of an intermetal dielectric (IMD) layer on the ILD layer; forming a second portion of the IMD layer on the first portion of the IMD layer; forming a metal layer on the second portion of the IMD layer; etching through the metal layer, through the second portion of the IMD layer, through the first portion of the IMD layer, and partially into the ILD layer to form a plurality of deep trench structures that are partially in the ILD layer; forming a passivation layer in a bottom and sidewalls of each of the plurality of deep trench structures; and forming a humidity sensing layer in the plurality of deep trench structures and above the passivation layer, wherein, in each of the plurality of deep trench structures, a bottom surface of the humidity sensing layer is above a portion of the ILD layer.
2. The method of claim 1, further comprising: etching the metal layer to form one or more electrodes and one or more electrode pads.
3. The method of claim 2, wherein the passivation layer is a first passivation layer; and wherein the method further comprises: forming a second passivation layer on the one or more electrodes; and wherein forming the first passivation layer in the bottom and the sidewalls of each of the plurality of deep trench structures comprises: forming the first passivation layer above the second

passivation layer and in the bottom and the sidewalls of each of the plurality of deep trench structures.
4. The method of claim 3, wherein the humidity sensing layer is formed above the one or more electrodes.
5. The method of claim 2, wherein the passivation layer is formed above the one or more electrodes.
6. The method of claim 5, wherein the humidity sensing layer is formed above the one or more electrodes.
7. The method of claim 1, wherein the plurality of deep trench structures are formed to have a depth in a range of 42,000 angstroms to 50,000 angstroms.
8. A method, comprising: forming an interlayer dielectric (ILD) layer on a substrate; forming a first portion of an intermetal dielectric (IMD) layer on the ILD layer; forming a second portion of the IMD layer on the first portion of the IMD layer; forming a metal layer on the second portion of the IMD layer; etching through the metal layer, through the second portion of the IMD layer, through the first portion of the IMD layer, and partially into the ILD layer to form a plurality of deep trench structures that are partially in the ILD layer; forming a passivation layer in a bottom and sidewalls of each of the plurality of deep trench structures; and forming a third dielectric layer in the plurality of deep trench structures and above the passivation layer, wherein, in each of the plurality of deep trench structures, a bottom surface of the third dielectric layer is above a portion of the ILD layer.
9. The method of claim 8, further comprising: etching the metal layer to form one or more electrodes.
10. The method of claim 9, wherein the passivation layer is a first passivation layer; and wherein the method further comprises: forming a second passivation layer on the one or more electrodes; and wherein forming the first passivation layer in the bottom and the sidewalls of each of the plurality of deep trench structures comprises: forming the first passivation layer above the second passivation layer and in the bottom and the sidewalls of each of the plurality of deep trench structures.
11. The method of claim 10, wherein the third dielectric layer is formed above the one or more electrodes.
12. The method of claim 9, wherein the passivation layer is formed above the one or more electrodes.
13. The method of claim 12, wherein the third dielectric layer is formed above the one or more electrodes.
14. The method of claim 8, wherein the plurality of deep trench structures are formed to have a depth in a range of 42,000 angstroms to 50,000 angstroms.
15. A method, comprising: forming an interlayer dielectric (ILD) layer; forming a first portion of an intermetal dielectric layer (IMD) above the ILD layer; forming a second portion of the IMD layer on the first portion of the IMD layer, wherein the first portion and the second portion of the IMD layer is formed in separate deposition operations; forming a metal layer above the second portion of the IMD layer; etching through the metal layer, through the second portion of the IMD layer, through the first portion of the IMD layer, and partially into the ILD layer to form a first deep trench structure that is partially in the ILD layer; etching through the metal layer, through the second portion of the IMD layer, through the first portion of the IMD layer, and partially into the ILD layer to form a second deep trench structure that is partially in the ILD layer; forming a passivation layer in a bottom and sidewalls of the first deep trench structure and the second deep trench structure; and forming a third dielectric layer in the first deep trench structure, in the second deep trench structure, above the passivation layer, and above the metal layer, wherein, in each of the first deep trench structure and the second deep trench structure, a bottom surface of the third dielectric layer is above a portion of the ILD layer.
16. The method of claim 15, further comprising: etching the metal layer to form a plurality of

electrodes.
17. The method of claim 16, wherein the passivation layer is a first passivation layer; and wherein the method further comprises: forming a second passivation layer on the plurality of electrodes; and wherein forming the first passivation layer in the bottom and the sidewalls of the first deep trench structure and the second deep trench structure comprises: forming the first passivation layer above the second passivation layer and in the bottom and the sidewalls of the first deep trench structure and the second deep trench structure.
18. The method of claim 17, wherein the third dielectric layer is formed above the plurality of electrodes.
19. The method of claim 16, wherein the passivation layer is formed above the plurality of electrodes.
20. The method of claim 19, wherein the third dielectric layer is formed above the plurality of electrodes.